

Ruofan Wu

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I am a third-year PhD candidate in CSE at the University of Michigan, advised by Prof. Mosharaf Chowdhury. My research focuses on energy-efficient ML systems and compilers, with two complementary goals: reducing energy consumption and maximizing performance under power constraints. I pursue these goals through system-level resource management, compiler optimizations, and GPU kernel scheduling.

EDUCATION

University of Michigan

Ann Arbor, MI, USA

Ph.D. candidate in Computer Science and Engineering, advised by Prof. Mosharaf Chowdhury

Sep 2024 - Present

Renmin University of China

Beijing, China

M.E. in Computer Application Technology, advised by Prof. Feng Zhang

Sep 2021 - Jun 2024

Honors & Awards: Scholarship (2023, 2022, 2021)

Renmin University of China

Beijing, China

B.E. in Data Science and Big Data Technology

Sep 2017 - Jun 2021

Honors & Awards: Excellent Bachelor Dissertation; Scholarship (2020, 2019); Sa-Shixuan Fund Scholarship (2019)

PUBLICATIONS

Energy-Efficient ML Systems

- **Kareus: joint reduction of dynamic and static energy in large model training**, Ruofan Wu, Jae-Won Chung, Mosharaf Chowdhury. **OSDI**, 2026.
- **The ML.ENERGY benchmark: toward automated inference energy measurement and optimization**, Jae-Won Chung, Jeff J. Ma, Ruofan Wu, Jiachen Liu, Oh Jun Kweon, Yuxuan Xia, Zhiyu Wu, Mosharaf Chowdhury. **NeurIPS Datasets and Benchmarks (Spotlight)**, 2025.
- **Where do the joules go? Diagnosing inference energy consumption**, Jae-Won Chung, Ruofan Wu, Jeff J. Ma, Mosharaf Chowdhury. Preprint, 2026.
- **Energy calculus: a compositional algebra of energy in computational systems**, Mosharaf Chowdhury, Jae-Won Chung, Jeff J. Ma, Nishil Talati, Ruofan Wu. Preprint, 2026.

ML Compilers and Serving Systems

- **TetriServe: efficient DiT serving for heterogeneous image generation**, Runyu Lu*, Shiqi He*, Wenxuan Tan, Shenggui Li, Ruofan Wu, Jeff J. Ma, Ang Chen, Mosharaf Chowdhury. **ASPLOS**, 2026.
- **PluS: highly efficient and expandable ML compiler with pluggable graph schedules**, Ruofan Wu, Zhen Zheng, Feng Zhang, Chuanjie Liu, Zaifeng Pan, Jidong Zhai, Xiaoyong Du. **USENIX ATC**, 2025.
- **RecFlex: enabling feature heterogeneity-aware optimization for deep recommendation models with flexible schedules**, Zaifeng Pan, Zhen Zheng, Feng Zhang, Bing Xie, Ruofan Wu, Shaden Smith, Chuanjie Liu, Olatunji Ruwase, Xiaoyong Du, Yufei Ding. **SC**, 2024.
- **RECom: a compiler approach to accelerating recommendation model inference with massive embedding columns**, Zaifeng Pan, Zhen Zheng, Feng Zhang, Ruofan Wu, Hao Liang, Dalin Wang, Xiafei Qiu, Junjie Bai, Wei Lin, Xiaoyong Du. **ASPLOS**, 2024.
- **ROLLER: fast and efficient tensor compilation for deep learning**, Hongyu Zhu, Ruofan Wu, Yijia Diao, Shanbin Ke, Haoyu Li, Chen Zhang, Jilong Xue, Lingxiao Ma, Yuqing Xia, Wei Cui, Fan Yang, Mao Yang, Lidong Zhou, Asaf Cidon, Gennady Pekhimenko. **OSDI**, 2022.

GPU Kernel Optimization

- **DREW: efficient Winograd CNN inference with deep reuse**, Ruofan Wu, Feng Zhang, Jia Wei Guan, Zhen Zheng, Xiaoyong Du, Xipeng Shen. **WWW/TheWebConf**, 2022.
- **Expanding the edge: enabling efficient Winograd CNN inference with deep reuse on edge device**, Feng Zhang, Ruofan Wu, Jiawei Guan, Zhen Zheng, Xiaoguang Guo, Xiao Zhang, Xiaoyong Du, Xipeng Shen. **IEEE TKDE**, 2023.
- **TREC: transient redundancy elimination-based convolution**, Jiawei Guan, Feng Zhang, Jiesong Liu, Hsin-Hsuan Sung, Ruofan Wu, Xiaoyong Du, Xipeng Shen. **NeurIPS**, 2022.
- **YuenyeungSpTRSV: a thread-level and warp-level fusion synchronization-free sparse triangular solve**, Jiya Su, Feng Zhang, Weifeng Liu, Bingsheng He, Ruofan Wu, Xiaoyong Du, Rujia Wang. **IEEE TPDS**, 2021.
- **CapelliniSpTRSV: a thread-level synchronization-free sparse triangular solve on GPUs**, Jiya Su, Feng Zhang, Weifeng Liu, Bingsheng He, Ruofan Wu, Xiaoyong Du, Rujia Wang. **ICPP**, 2020.

EXPERIENCE

NVIDIA

PhD Research Intern at ADLR & Compute Arch

Mentors: Dr. Ahmet Inci, Dr. Bhargava Gopireddy, and Dr. Deepak Narayanan
Energy-efficient large-model training with Megatron-LM.

Santa Clara, CA, USA

May 2026 - Aug 2026

Microsoft Software Technology Center Asia

Research Intern at Bing, mentored by Dr. Zhen Zheng

Extensible ML compilation (**PluS**) and low-precision GPU kernel optimization.

Beijing, China

Oct 2023 - Jun 2024

Alibaba Cloud

Research Intern at Platform of Artificial Intelligence, mentored by Dr. Zhen Zheng

Compiler optimizations for efficient recommendation model inference.

Hangzhou, China

Jan 2022 - Sep 2023

Microsoft Research Asia

Research Intern at Systems Research Group, mentored by Dr. Jilong Xue and Dr. Fan Yang

Efficient tensor compilation through hardware-aware tiling (**ROLLER**).

Beijing, China

Apr 2021 - Dec 2021

North Carolina State University

Research Collaborator with PICTURE Research Group, mentored by Prof. Xipeng Shen

Efficient CNN inference through redundancy reduction (**DREW**).

Remote

Nov 2019 - Mar 2021

TALKS, TEACHING, AND SERVICE

- **Kareus**, invited talk at the PICASSO Lab seminar, UC San Diego (Jan 2026)
- **Energy and Power as First-Class ML Design Metrics**, tutorial at NeurIPS 2025
- **Energy and Power as First-Class Resources across the LLM Lifecycle**, invited lecture for CSE 585 (Nov 2025)
- **Academic Service**: Reviewer for IEEE TPDS (Dec 2023)
- **Teaching Assistant**: Parallel Architecture and Programming, Renmin University of China (Fall 2022)